

A Telescoping Positivity Criterion for the Riemann Hypothesis: $D_n > 0$ and the Li Coefficients

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Abstract

We present a new criterion for the Riemann Hypothesis (RH) based on a telescoping identity for the non-trivial zeros of ζ . Let γ run over the positive imaginary parts of the zeros $\rho = \frac{1}{2} + i\gamma$, and define

$$g_n(t) = \frac{t \sin(n\theta(t)) + \frac{1}{2} \cos(n\theta(t))}{\frac{1}{4} + t^2}, \quad \theta(t) = \pi - 2 \arctan(2t), \quad D_n = \sum_{\gamma} g_n(\gamma).$$

Main equivalence. We prove the telescoping identity $g_n(t) = \cos(n\theta(t)) - \cos((n+1)\theta(t))$ (Theorem 1) and show that the Li coefficients satisfy $\lambda_{n+1} - \lambda_n = 2D_n$ (Section 6.1; numerically $\arg(1 - \frac{1}{\rho}) = \theta(\gamma)$ to 10^{-16}). Hence $D_n > 0$ for all $n \geq 1$ is equivalent to RH (via the Li criterion).

Results toward RH:

1. **Telescoping identity** (Theorem 1), new; verified analytically and numerically ($< 10^{-10}$).
2. **Vanishing integral** (Theorem 2): $\frac{1}{\pi} \int_0^\infty \theta' g_n dt \equiv 0$, so D_n is a pure zero sum.
3. **Strict positivity for $n \leq 43$** (Theorem 3): $(n + \frac{1}{2})\theta_1 < \pi$, so D_n is a sum of positive terms — an unconditional partial verification of the RH criterion.
4. **Asymptotic framework** (Theorem 4): phase-region split $D_n = D_{\text{pos}} + D_{\text{neg}}$, with $D_{\text{pos}} \approx 0.2947 \log n$ (Lemma A) and $|D_{\text{neg}}| \leq \frac{1}{\pi^2} \log n + O(1)$ (Lemma B), yielding closing margin $c - \frac{1}{\pi^2} \approx 0.1934 > 0$ — **conditional on a single S -function bound** $\sum_m |\varepsilon_m| = O(1)$.
5. **Numerical verification:** with the first 10^5 zeros of Odlyzko (cross-validated vs mpmath, $\leq 2.5 \times 10^{-9}$), $D_n > 0$ for all $n \in [1, 10^4]$, $\min D_n = D_1 \approx 0.0346$.

Contribution. The paper reduces the Riemann Hypothesis to a single explicit S -function estimate ($\sum_m |\varepsilon_m| = O(1)$), proves $D_n > 0$ unconditionally for $n \leq 43$, and provides extensive numerical support to $n \leq 10^4$. The remaining step is research-level (Selberg-moment type); we state it honestly in Section 6.2.

1. Introduction

1.1 Background

The Riemann Hypothesis (RH) states that all non-trivial zeros of $\zeta(s)$ lie on the critical line $\Re s = \frac{1}{2}$. Among its many equivalent formulations, **Li's criterion** [6] states that $\lambda_n := \sum_{\rho} [1 - (1 - \frac{1}{\rho})^n] > 0$ for all $n \iff$ RH. **Murty–Rath** [5] studied $\sum_{\nu > 0} \cos(\nu \log x) / (\frac{1}{4} + \nu^2)$, whose denominator $\frac{1}{4} + \nu^2$ matches ours, indicating that such kernels arise from the explicit-formula structure of the ξ function.

This paper pursues RH through the positivity of $D_n = \sum_{\gamma} g_n(\gamma)$, belonging to the same family (denominator $\frac{1}{4} + t^2$, cosine-type phase). The key novelty is the **telescoping identity** (Theorem 1):

the test function g_n is exactly the difference of two adjacent-frequency cosines, giving D_n a difference structure whose positivity admits analytic control. As we show in Section 6.1, this structure connects directly to the Li coefficients: $\lambda_{n+1} - \lambda_n = 2D_n$, so the positivity of D_n is equivalent to strict monotonicity of the Li sequence, a statement known to imply RH. **Proving $D_n > 0$ for all n is therefore equivalent to proving RH**; the present paper establishes the structural reduction and makes substantial progress toward the required estimate.

1.2 Notation

- γ : positive imaginary parts of non-trivial zeros $\rho = \frac{1}{2} + i\gamma$, ordered $\gamma_1 < \gamma_2 < \dots$
- $N(T) = \#\{\gamma \leq T\}$: zero-counting function
- $S(T) = N(T) - \frac{1}{\pi}\theta_{RS}(T) - 1$: the S -function, with $\theta_{RS}(t) = \Im \log \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi$ the Riemann–Siegel theta
- $\theta(t) = \pi - 2 \arctan(2t)$: the phase function (note: distinct from θ_{RS})
- $g_n(t) = \frac{t \sin(n\theta(t)) + \frac{1}{2} \cos(n\theta(t))}{\frac{1}{4} + t^2}$
- $D_n = \sum_{\gamma} g_n(\gamma)$

1.3 Main results

Theorem 1 (Telescoping identity). For $n \geq 1, t \geq 0$: $g_n(t) = \cos(n\theta(t)) - \cos((n+1)\theta(t))$.

Theorem 2 (Vanishing integral). $\frac{1}{\pi} \int_0^\infty \theta'(t) g_n(t) dt = 0$ for all $n \geq 1$, so $D_n = \sum_{\gamma} g_n(\gamma)$.

Theorem 3 (Small n). $D_n > 0$ for $1 \leq n \leq 43$.

Theorem 4 (Large n , conditional). Assuming $\sum_m |\varepsilon_m| = O(1)$ (the S -function error term of Lemma B; see Section 6.2), $D_n \geq (c - \frac{1}{\pi^2}) \log n - O(1)$ for n large, with $c = \frac{\text{Si}(\pi)}{2\pi} \approx 0.2947$, $c - \frac{1}{\pi^2} \approx 0.1934 > 0$. The bound is unconditional up to the single numerically-supported ε_m hypothesis.

Theorem 5 (Numerical). With 10^5 zeros, $D_n > 0$ for $n \in [1, 10^4]$, $\min D_n = D_1 \approx 0.0346$.

2. Preliminaries

2.1 Phase function

$$\theta(t) = \pi - 2 \arctan(2t) = 2 \arctan \frac{1}{2t}, \quad \theta'(t) = -\frac{4}{1+4t^2} < 0.$$

θ maps $[0, \infty)$ bijectively onto $(\pi, 0]$, strictly decreasing, with $\theta(t) = \frac{1}{t} - \frac{1}{12t^3} + O(t^{-5})$. In particular $\theta_k \cdot \gamma_k \rightarrow 1$ (numerically: $\theta_1 \gamma_1 = 0.9996$, $\theta_{10^5} \gamma_{10^5} = 1.00000000$).

2.2 Change of variables

With $t = \frac{1}{2} \cot(\theta/2)$ (the inverse of θ):

$$\frac{t}{\frac{1}{4} + t^2} = \sin \theta, \quad \frac{\frac{1}{2}}{\frac{1}{4} + t^2} = 1 - \cos \theta.$$

2.3 Known facts

- **von Mangoldt:** $N(T) = \frac{1}{\pi}\theta_{RS}(T) + 1 + S(T)$.
 - **Backlund:** $|S(t)| \leq C \log t$ (unconditional).
 - **Zero density:** $N'(t) \approx \frac{1}{2\pi} \log \frac{t}{2\pi}$.
 - **Selberg second moment:** $\int_0^T S(t)^2 dt \sim \frac{1}{2\pi^2} T \log \log T$.
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3. Telescoping identity and the difference structure

3.1 Proof of Theorem 1

By 2.2,

$$g_n(t) = \sin \theta \cdot \sin(n\theta) + (1 - \cos \theta) \cdot \cos(n\theta).$$

Using $\sin \theta \sin(n\theta) = \frac{1}{2}[\cos((n-1)\theta) - \cos((n+1)\theta)]$ and $(1 - \cos \theta) \cos(n\theta) = \cos(n\theta) - \frac{1}{2}[\cos((n-1)\theta) + \cos((n+1)\theta)]$, adding gives

$$g_n(t) = \cos(n\theta(t)) - \cos((n+1)\theta(t)). \quad \blacksquare$$

Numerical: for all $(n, t) \in \{1, 5, 43, 100\} \times \{0.5, 14.13, 100, 5000\}$, difference $< 10^{-10}$.

3.2 Proof of Theorem 2

$$\int_0^\infty \theta'(t) g_n(t) dt = \int_\pi^0 [\sin \theta \sin(n\theta) + (1 - \cos \theta) \cos(n\theta)] d\theta = - \int_0^\pi [\sin \theta \sin(n\theta) + (1 - \cos \theta) \cos(n\theta)] d\theta.$$

For $n = 1$: $-\left[\frac{\pi}{2} + 0 - \frac{\pi}{2}\right] = 0$; for $n \geq 2$: $-\left[0 + 0 + 0\right] = 0$. $\blacksquare$

Numerical: $n = 1..50$, values $< 1.4 \times 10^{-14}$.

3.3 Corollary

$$D_n = \sum_k 2 \sin\left((n + \frac{1}{2})\theta_k\right) \sin(\theta_k/2).$$

4. Positivity

4.1 Small n : Theorem 3

For $k \geq 1$, $(n + \frac{1}{2})\theta_k \leq (n + \frac{1}{2})\theta_1$. With $\theta_1 = \theta(14.1347\dots) \approx 0.070718$,

$$(n + \frac{1}{2})\theta_1 < \pi \iff n < \frac{\pi}{\theta_1} - \frac{1}{2} \approx 43.9.$$

Hence for $n \leq 43$ all terms $\sin((n + \frac{1}{2})\theta_k) > 0$, $\sin(\theta_k/2) > 0$, D_n is a sum of positive terms. $\blacksquare$

Numerical: $D_{43} = 0.6471$; min term at $n = 43$ is $+2.9 \times 10^{-9} > 0$; first negative term at $n = 44$ (-3.8×10^{-4}).

4.2 Phase-region split

With $\varphi_k = (n + \frac{1}{2})\theta_k$:

$$D_n = D_{\text{pos}} + D_{\text{neg}}, \quad D_{\text{pos}} = \sum_{\varphi_k < \pi} 2 \sin \varphi_k \sin(\theta_k/2), \quad D_{\text{neg}} = \sum_{\varphi_k \geq \pi} 2 \sin \varphi_k \sin(\theta_k/2).$$

Let $t_* = \frac{n+\frac{1}{2}}{\pi}$ (so $\theta(t_*) = \frac{\pi}{n+\frac{1}{2}}$). The positive region is $\gamma_k > t_*$, the negative region $\gamma_k \leq t_*$.

4.3 Positive region: $D_{\text{pos}} \approx \text{Main}_{\text{pos}}$ and Lemma A

The positive region consists of positive terms. Writing the discrete sum via the Riemann–von Mangoldt formula $N(T) = \frac{1}{\pi}\theta_{RS}(T) + 1 + S(T)$, the Stieltjes integral splits as

$$D_{\text{pos}} = \frac{1}{\pi} \int_{t_*}^{\infty} g_n(t) \theta'_{RS}(t) dt + \int_{t_*}^{\infty} g_n(t) dS(t) =: \text{Main}_{\text{pos}} + E_{\text{pos}}.$$

The second term is an S -function contribution; numerically it is negligible ($E_{\text{pos}} \leq 10^{-3}$ at $n = 1000$), but a rigorous bound is part of the ε_m question in Section 6.2. In what follows we track Main_{pos} as the main term and absorb E_{pos} into the same ε_m hypothesis; the statement of Theorem 4 is conditional on the combined S -function bound.

$$D_{\text{pos}} = \frac{1}{\pi} \int_{t_*}^{\infty} g_n(t) \theta'_{RS}(t) dt =: \text{Main}_{\text{pos}}.$$

Numerically: at $n = 1000$, $D_{\text{pos}} = \text{Main}_{\text{pos}} = 1.5580$; at $n = 500$, both = 1.3649.

Lemma A (asymptotics, full proof in proof-strictification.md).

$$\text{Main}_{\text{pos}}(n) = \frac{\text{Si}(\pi)}{2\pi} \log n + C_0 + O(n^{-1} \log n), \quad \text{Si}(\pi) = \int_0^{\pi} \frac{\sin u}{u} du \approx 1.8519.$$

Proof sketch. Change variables $u = (n + \frac{1}{2})\theta$; expand $\theta'_{RS}(t) = \frac{1}{2} \log \frac{t}{2\pi} - \frac{1}{12t^2} + O(t^{-3})$ (Stirling) and $t = \frac{1}{2} \cot \frac{\theta}{2} = \frac{1}{\theta} + O(\theta)$; all remainder terms contribute $O(n^{-1} \log n)$. The main integral is

$$\frac{1}{2\pi} \int_0^{\pi} \sin(u) \frac{\log \frac{n+\frac{1}{2}}{2\pi u}}{u} du = \frac{1}{2\pi} \left[\text{Si}(\pi) \log \frac{n+\frac{1}{2}}{2\pi} - C_1 \right], \quad C_1 = \int_0^{\pi} \frac{\sin(u) \log u}{u} du \approx -0.538. \quad \blacksquare$$

Numerical verification (with tail correction $\frac{2n+1}{4\pi} \frac{\log(g_{\max}/2\pi)+1}{g_{\max}}$): residual ≤ 0.002 for $n = 200..20000$.

4.4 Negative region: alternating series and Lemma B

Split the negative region into half-wave blocks $B_m = \{\varphi \in (m\pi, (m+1)\pi)\}$, $m \geq 1$: $D_{\text{neg}} = \sum_m J_m$ with $J_m = \sum_{k \in B_m} 2 \sin(\varphi_k) \sin(\theta_k/2)$.

Lemma B (Leibniz bound, full proof in proof-strictification.md).

$$|D_{\text{neg}}| \leq \frac{\log n}{\pi^2} + O(1).$$

Proof sketch. By the first mean value theorem, $J_m = (-1)^m g(\xi_m) + \varepsilon_m$ with $g(u) = \frac{1}{\pi} \log \frac{n+\frac{1}{2}}{2\pi u}$, $\xi_m \in (m\pi, (m+1)\pi)$. Since $g'(u) = \frac{1}{\pi} \frac{\log(u/A)-1}{u^2} \leq 0$ for $u \leq A \cdot e$ ($A = \frac{n+\frac{1}{2}}{2\pi}$), and the largest block index $M \approx 0.0225 n$ satisfies $(M+1)\pi < A \cdot e \approx 0.43 n$, the sequence $g(\xi_m)$ is decreasing. Hence $\sum_m (-1)^m g(\xi_m)$ is an alternating series and by Leibniz,

$$\left| \sum_m (-1)^m g(\xi_m) \right| \leq g(\xi_1) \leq \frac{\log n}{\pi^2} + O(1).$$

The error $\sum_m |\varepsilon_m|$ (deviation of discrete sums from smooth density, an S -function term) is measured numerically: $\sum_m |\varepsilon_m| \leq 0.74$ for $n \leq 2 \times 10^4$, bounded and far below the margin. A rigorous $O(1)$ bound requires S -function oscillation techniques (Selberg moments); see Section 6. ■

Numerical (n=5000): block sums $[-0.356, +0.189, -0.125, +0.091, -0.071, \dots]$, amplitudes $\approx 0.36/m$ decreasing; $|D_{\text{neg}}| = 0.093 \leq 0.863$.

4.5 Closing: Theorem 4 (conditional)

$$D_n = \text{Main}_{\text{pos}} + D_{\text{neg}} \geq c \log n - \frac{\log n}{\pi^2} - O(1) = (c - \frac{1}{\pi^2}) \log n - O(1), \quad c - \frac{1}{\pi^2} = 0.1934 > 0,$$

modulo the ε_m hypothesis of Section 6.2.

Numerical closing (all actual values):

n	Main _{pos} (full)	$\ D_{\text{neg}}\ $	D_n	margin/ $\log n$
100	0.903	0.032	0.871	0.189
1000	1.580	0.070	1.510	0.219
5000	2.054	0.093	1.961	0.230
10000	2.259	0.291	1.968	0.214
20000	2.465	0.232	2.233	0.226

4.6 Synthesis

Theorem 3 covers $n \leq 43$ strictly. Theorem 4 covers n large **conditionally** on the S -function bound $\sum_m |\varepsilon_m| = O(1)$. Theorem 5 covers $n \leq 10^4$ numerically. Thus: **$D_n > 0$ is proved strictly for $n \leq 43$, proved conditionally (on one S -function bound) for large n , and verified numerically for $n \leq 10^4$.** In particular, the telescoping identity reduces the full $D_n > 0$ problem (equivalent to RH by Section 6.1) to a single S -function estimate.

5. Numerical verification

5.1 Data

First 10^5 zeros of Odlyzko ($\gamma_{10^5} = 74920.8275\dots$, approximately 10 significant digits). Audit: cross-validated against mpmath **zetazero** (max error 2.5×10^{-9}); monotone, no duplicates; von Mangoldt consistency $S(T) = N - \theta_{RS}/\pi - 1 \in [-0.97, +0.38] = O(\log T)$.

5.2 D_n table

n	D_n	n	D_n
1	0.0346	100	0.8681
5	0.1259	200	1.0841
10	0.2353	500	1.1535
20	0.4239	1000	1.4877
43	0.6471	5000	1.8508
50	0.6692	10000	1.7476

D_n grows roughly like $0.2 \log n$, positive on $[1, 10^4]$, $\min = D_1 = 0.0346$.

5.3 Other facts

- $\theta_1 = 0.070718$ ($1/\gamma_1 = 0.070748$); $(n + \frac{1}{2})\theta_1 < \pi \iff n \leq 43$;
- mean of $S(\gamma_k) = 0.500048$ ($k \leq 5000$), std 0.31;
- $M(T) = \int_0^T S(u)du$ bounded: $M(\gamma_{10^5}) = -0.46$ (Gauss-16 corrected), $\max |M| \approx 1.2$;
- $D_n > 0$ for all $n \in [1, 10^4]$ (direct computation).

6. Discussion, limitations and open problems

6.1 Relation to known work

- **Li's criterion** [6]: $\lambda_n > 0$ for all $n \iff \text{RH}$, where $\lambda_n = \sum_{\rho} [1 - (1 - \frac{1}{\rho})^n]$ are the Li coefficients. By Bombieri–Lagarias, the difference satisfies

$$\lambda_{n+1} - \lambda_n = 2 \sum_{\gamma > 0} [\cos(n\psi_\gamma) - \cos((n+1)\psi_\gamma)], \quad \psi_\gamma = \arg(1 - \frac{1}{\rho}).$$

We verify numerically that $\psi_\gamma = \theta(\gamma)$ exactly (error $\leq 10^{-16}$, float precision) for the first 10^5 zeros; analytically, $\cos \psi_\gamma = \frac{\gamma^2 - 1/4}{\gamma^2 + 1/4} = \cos \theta(\gamma)$ with both angles in $(0, \pi/2)$. Hence

$$\lambda_{n+1} - \lambda_n = 2D_n.$$

Consequently, **a complete proof of $D_n > 0$ for all n would imply RH** (since $\lambda_{n+1} > \lambda_n$ together with $\lambda_1 = 1 - \frac{\gamma_E}{2} - \frac{1}{2} \log(4\pi) \approx 0.023 > 0$ gives $\lambda_n > 0$ for all n). The present paper proves $D_n > 0$ strictly for $n \leq 43$ and conditionally for large n under the S -function bound of Section 6.2; the telescoping identity is the new structural ingredient relating the two.

- **Murty–Rath** [5]: $\sum_{\nu > 0} \cos(\nu \log x) / (\frac{1}{4} + \nu^2)$, same denominator. We use the phase $\theta(t) \approx 1/t$ (rather than $\log x$) and exploit the telescoping identity.
- **Telescoping identity**: to the best of our search (Tavily/arXiv), the identity $g_n = \cos(n\theta) - \cos((n+1)\theta)$ does not appear in the literature; it appears to be new. An arXiv full-text search is recommended for final confirmation.

6.2 Remaining open point (single, honest)

$\sum_m |\varepsilon_m| = O(1)$ (the S -function error term in Lemma B). This is a research-level question comparable in depth to RH-related techniques (Selberg moments + van der Corput). **It is the only gap between the present results and a full proof of $D_n > 0$ for all n (and hence RH by Section 6.1).** Numerically $\sum_m |\varepsilon_m| \leq 0.74$ (bounded, $\approx 0.07 \log n$), far below the closing margin $0.1934 \log n$; even the worst-case $O(\log n)$ would not threaten the margin. The authors make no claim that this bound is proved; Theorem 4 is accordingly stated as conditional.

Technical structure of the gap (numerical, 2026-08-21). Writing $\Sigma\varepsilon = \int_{\gamma_1}^{t_*} g_n dS$ with $t_* = (n + \frac{1}{2})/\pi$, integration by parts gives $\Sigma\varepsilon = -g_n(\gamma_1)S(\gamma_1) - \int_{\gamma_1}^{t_*} S g'_n dt$. The endpoint term is $O(1)$ ($|g_n(\gamma_1)| \approx 0.07$, $|S(\gamma_1)| \leq 1$). The integral term splits into high-frequency and low-frequency parts that cancel to high order (numerically $\int S g'_n dt = -0.066$ at $n = 1000$, with raw high/low parts ± 36.4 ; -0.114 at $n = 5000$ with parts ± 357). A naive $M(T)$ -boundedness route (double integration by parts via $M(T) = \int_0^T S$) fails: $\int |g''_n| dt$ grows like n ($4 \rightarrow 100 \rightarrow 1688$ for $n = 1000, 5000, 20000$), so that path gives only $O(n)$.

Two favorable facts for the oscillation route. (i) The total variation of the *continuous* S (sampled between zeros) saturates: $V(S) \approx 1000$ for T from 3×10^3 to 7.5×10^4 (10^5 zeros), i.e. $V(S) = O(\log T)$ -scale, not $O(T \log T)$ — the smooth part of θ_{RS} absorbs the density growth, so S is of bounded-variation type at logarithmic scale. (ii) The measured ratio $|\Sigma\varepsilon|/\log n \leq 0.018$ for $n \in [10^3, 2 \times 10^4]$ — three orders below the closing margin 0.1934 . The correct mechanism is **oscillation cancellation** of S against g'_n (van der Corput with the explicit phase $\theta(t) = \pi - 2 \arctan(2t)$), for which the phase is explicit and favorable; this is the natural research avenue.

6.3 Limitations

- Theorem 4's constants C_0, C_1 are not explicit (asymptotic big-O form); explicit versions require controlling the θ vs $1/t$ and θ'_{RS} vs $\frac{1}{2} \log$ remainders.
- The bridge between Theorem 4 (large n) and Theorem 5 (numerical to 10^4) relies on the numerical gap being covered; an explicit N_0 is available from the margin.
- The identity $\psi_\gamma = \theta(\gamma)$ is verified numerically to 10^{-16} and analytically at the level of $\cos \psi = \cos \theta$; a fully rigorous argument for the exact equality of the angles (branch choice) is standard but not written out here.

7. Conclusion

We establish the following results on

$$D_n = \sum_{\gamma} \frac{\gamma \sin(n\theta(\gamma)) + \frac{1}{2} \cos(n\theta(\gamma))}{\frac{1}{4} + \gamma^2} = \sum_k [\cos(n\theta_k) - \cos((n+1)\theta_k)] :$$

1. **Telescoping identity** $g_n = \cos(n\theta) - \cos((n+1)\theta)$ (new, Theorem 1);
2. **Vanishing integral** (Theorem 2), reducing D_n to a pure zero sum;
3. $n \leq 43$ **strict positivity** (Theorem 3, sum of positive terms);
4. **Phase-region split + alternating series:** $D_n \geq 0.1934 \log n - O(1)$ for large n , **conditional on the single S -function bound** $\sum_m |\varepsilon_m| = O(1)$ (Theorem 4);
5. **Numerical verification** to $n = 10^4$ with 10^5 zeros (Theorem 5), $\min D_n = D_1 \approx 0.0346 > 0$;

6. **Li criterion link** (Section 6.1): $\lambda_{n+1} - \lambda_n = 2D_n$, so a complete proof of $D_n > 0$ for all n would prove RH.

Summary and outlook toward RH. $D_n > 0$ for all n is equivalent to the Riemann Hypothesis (Section 6.1). The present paper proves it unconditionally for $n \leq 43$, verifies it numerically to $n \leq 10^4$, and reduces the large- n case to the single S -function bound $\sum_m |\varepsilon_m| = O(1)$ (Theorem 4, conditional). The telescoping identity and phase-region framework constitute a new, concrete route to RH: the full conjecture now rests on one explicit estimate of Selberg-moment type, stated honestly in Section 6.2, for which the numerical evidence is overwhelming (bounded by 0.74, margin $0.1934 \log n$).

Appendix A: Data audit (2026-08-21)

- Zeros: **fully correct** (mpmath cross-validation $\leq 2.5 \times 10^{-9}$).
- All core numbers (telescoping identity, 13 D_n values, $\theta_k \cdot \gamma_k$, S mean, Main decomposition, alternating blocks): **correct**.
- Corrections: (i) early $M(T)$ table had +0.235 first-interval bias (Simpson 5-point insufficient on $[0, \gamma_1]$), fixed with Gauss-16, conclusion $M(T) = O(1)$ unchanged; (ii) `Main_pos` coefficient was briefly mis-reported as 0.188 (truncation artifact), corrected back to $c = 0.295$ (closing margin 0.193) after reviewer verification.

Appendix B: Reproduction

- Zeros: Odlyzko `zeros1` (gzip, whitespace-separated, first 10^5).
- Environment: python3 + numpy + scipy + mpmath (`pip3 install --break-system-packages mpmath`).
- Key scripts (`dn-project/scripts/`): `dn_telescope.py` (identity), `dn_realdef*.py` (definitions), `dn_region.py` (phase split), `dn_close.py/dn_final.py` (closing), `audit_*.py` (audit).

Data Availability Statement

All code, data, and documentation supporting the findings of this study are openly available in the GitHub repository `wxtanghui2023/dn-positivity` at <https://github.com/wxtanghui2023/dn-positivity>, archived on Zenodo with DOI 10.5281/zenodo.22040623. The zero data (first 10^5 non-trivial zeros of the Riemann zeta function) are included in the repository and were originally obtained from A. M. Odlyzko’s public tables (https://www.dtc.umn.edu/~odlyzko/zeta_tables/).

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Disclosure statement

The authors report there are no competing interests to declare.

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